Neil Ashton

Distinguished Engineer, NVIDIA

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Computational engineering researcher and technical leader working at the intersection of scientific machine learning, high-fidelity simulation, foundation models, and agentic AI for verifiable and traceable engineering workflows. Experience spans research and product direction, open scientific datasets, international benchmarking programmes, high-performance computing, and industrial deployment.

Experience

Jan 2025–present

Distinguished Engineer

NVIDIA | United Kingdom

- Set research and product direction for accelerated computational engineering, AI physics, simulation, and engineering AI.
- Lead work on foundation models, scientific machine learning, and agentic architectures, with emphasis on tool orchestration, physical verification, provenance, uncertainty, failure recovery, and human oversight.
- Work across research, engineering, product, and external partners to translate advances into dependable computational-engineering systems.

Feb 2020–Jan 2025

Worldwide Technical Lead for Computer-Aided Engineering

Amazon Web Services

- Led worldwide technical strategy for CAE across high-performance computing, machine learning, cloud services, and partner technologies.
- Founded and led a cross-Amazon machine-learning-for-CAE initiative, connecting product development, research, and industrial adoption.
- Provided technical leadership for open, high-fidelity datasets including AhmedML, WindsorML, and DrivAerML.

Jul 2016–Jan 2020

Senior Researcher

Osney Thermofluids Institute, Department of Engineering Science, University of Oxford

- Led vehicle-aerodynamics research spanning high-fidelity CFD, turbulence modelling, multiphysics, and high-performance computing.
- Supervised doctoral research and led collaborations with academic, government, and industrial partners.
- Founded Oxford University Racing and chaired the 10th International Conference on Computational Fluid Dynamics.

Jul 2017–Dec 2019

Consultant

Formula 1 Management

- Provided CFD and HPC expertise to the motorsport team, co-authored the CFD/HPC sections of proposed FIA Sporting Regulations, reviewed Technical Regulations, and helped establish a cloud-based OpenFOAM process.

Jul 2016–Dec 2019

Consultant

British Cycling

- Conducted the high-fidelity CFD programme for the Great Britain track bicycle developed for the Tokyo 2020 Olympic Games, including more than 1,000 transient simulations.

Jan 2016–Jul 2017

Consultant

Williams Advanced Engineering

- Established CFD and aerodynamic methods for Formula 3, Le Mans, and touring-car programmes.

Jan–Jul 2016

Visiting Scholar

Advanced Supercomputing Division, NASA Ames Research Center

- Conducted large-scale computational aerodynamics research for transonic and supersonic aircraft with the Applied Modeling and Simulation group.

2013–2015

EPSRC Impact Acceleration Research Associate

University of Manchester

- Developed turbulence models for automotive and aerospace applications and contributed to the EU FP7 Go4Hybrid programme.
- Undertook a one-month research visit to NASA Ames Research Center in February 2015.

2012–2013

CFD Engineer

Lotus F1 Team

- Developed computational-fluid-dynamics and high-performance-computing methods within the Formula 1 aerodynamics group.

Education

2008–2013
PhD, Aerospace Engineering
University of Manchester

Thesis: Development, Implementation and Testing of an Alternative DDES Formulation Based on Elliptic Relaxation. EPSRC funded.

2004–2008
MEng (Hons), Aerospace Engineering – First Class
University of Manchester

Professional standing

Fellow, Institution of Mechanical Engineers	Since 2020
Chartered Engineer, Institution of Mechanical Engineers	Since 2013
Senior Member, American Institute of Aeronautics and Astronautics	Since 2014
Member, NAFEMS CFD Working Group	Since 2014
Expert reviewer, European Commission	Since 2017
Member, IMechE Technical Strategy Board and Education Advisory Group	2017–2021

Research leadership and public scholarship

2019–present
Automotive CFD Prediction Workshop (AutoCFD)
Founder and organiser

Founded the international series for open test cases, reproducible comparisons, and shared automotive-CFD research priorities.

Current
AIAA CFD High-Lift Prediction Workshop (HLPW)
Organising Committee and AI/ML Technical Focus Group Lead

Directs aerospace benchmarking and leads machine-learning data and evaluation activities.

Current
CAE ML Datasets
Research lead and collaborator

Leads the open AhmedML, WindsorML, DrivAerML, and HiLiftAeroML collection of more than 3,100 high-fidelity simulation samples.

2024–present
The Neil Ashton Podcast
Founder and host

Long-form conversations with research and engineering leaders in AI, scientific computing, simulation, and aerodynamics.

Selected publications

1. 2026 Neil Ashton et al. HiLiftAeroML: High-Fidelity Computational Fluid Dynamics Dataset for High-Lift Aircraft Aerodynamics. *arXiv preprint*.

2. 2025 Neil Ashton, Johannes Brandstetter, and Siddhartha Mishra. Fluid Intelligence: A Forward Look on AI Foundation Models in Computational Fluid Dynamics. *arXiv preprint*.

3. 2024 Neil Ashton et al. WindsorML: High-Fidelity Computational Fluid Dynamics Dataset for Automotive Aerodynamics. *Advances in Neural Information Processing Systems 37, 37823–37835*.

4. 2024 Neil Ashton, Paul Batten, Andrew Cary, and Kevin Holst. Summary of the 4th High-Lift Prediction Workshop Hybrid RANS/LES Technology Focus Group. *Journal of Aircraft 61(1), 86–115*.

5. 2023 Neil Ashton and William Van Noordt. Overview and Summary of the First Automotive CFD Prediction Workshop: DrivAer Model. *SAE International Journal of Commercial Vehicles 16(1), 61–85*.

6. 2021 Jamil Appa, Mike Turner, and Neil Ashton. Performance of CPU and GPU HPC Architectures for Off-Design Aircraft Simulations. *AIAA SciTech 2021*.

7. 2016 Neil Ashton, Alastair West, Sylvain Lardeau, and Alistair Revell. Assessment of RANS and DES Methods for Realistic Automotive Models. *Computers & Fluids 128, 1–15*.

8. 2013 Neil Ashton, Alistair Revell, Robert Prosser, and Juan Uribe. Development of an Alternative Delayed Detached-Eddy Simulation Formulation Based on Elliptic Relaxation. *AIAA Journal 51(2), 513–519*.

Selected invited talks

Jul 2026	Sponsored keynote	**SIGGRAPH 2026** NVIDIA - Next Era of Graphics - Neural Rendering, World Models, and Simulation
May 2026	Invited seminar	**University of Oxford Mathematical Institute** Towards a Foundation Model for Computational Engineering: Opportunities, Challenges, and Novel Scaling Laws
Apr 2026	Invited seminar	**NASA Ames Seminar Series** Perspectives on Foundation Models and Agentic Architectures for Computational Fluid Dynamics
May 2025	Invited talk	**NASA High-Fidelity CFD Workshop** Enabling Industrially Relevant High-Fidelity CFD and AI Surrogate Models for External Aerodynamics
Apr 2023	Invited keynote	**22nd International Computational Fluids Conference** The Role of Cloud Computing, Machine Learning, and HPC in the Advancement of High-Fidelity CFD for Industry
Mar 2023	Invited seminar	**MIT ACDL Seminar Series** The Role of Cloud Computing, Machine Learning, and HPC in the Advancement of High-Fidelity CFD for Industry